

Carolina Del Pilar Díaz

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Summary

Enterprise AI Solution Architect & Staff Statistician (Economist) with 20+ years of experience spanning applied research, statistical modeling, and AI/ML engineering for global enterprises in Fintech, AgTech, and Healthcare. 10+ years leading statistical research and experimental design (INTA), followed by 6+ years architecting production Machine Learning, Deep Learning, and Multi-Agent GenAI systems (RAG, MCP) for Banco Galicia, ICBC, Banco Macro, and Aerolíneas Argentinas. Combines rigorous causal inference and econometrics with hands-on MLOps and cloud architecture (AWS, Azure, Data-bricks).

Experience

IT Patagonia (Consultant) , Staff AI Engineer / Architect	Remote, Argentina June 2024 – present
- Enterprise Architecture: Designed and led a proprietary Multi-Agent Framework (Lang-Graph, RAG, MCP), the production blueprint for autonomous AI applications at Banco Galicia and ICBC. - Agnostic System Design: Built provider-agnostic LLM architectures dynamically routing between Gemini, GPT, and Claude via n8n orchestration on hybrid cloud infrastructure. - AI Governance & XAI: Embedded statistical validation and SHAP-based interpretability layers into agent reasoning loops to prevent hallucinations and meet regulatory compliance.	
ITP Patagonia - ICBC , Senior Data Scientist / Technical Lead	Remote, Argentina June 2024 – July 2025
- Developed predictive models for payroll clients and mutual fund investment using AWS SageMaker, AutoML, Neural Networks, and XGBoost. - Built anomaly and fraud detection models (Autoencoder Neural Networks, Isolation Forest, Decision Trees, Time Series Analysis) with SHAP interpretability on Azure and AWS Fraud Detector. - Delivered customer segmentation (RFM) on ~1M monthly records using PySpark, Sage-Maker Canvas, and Amazon Athena.	
Nubiral S.A. , Senior AI Consultant / COE Expert	Argentina Oct 2023 – May 2024
- Integrated AWS Rekognition, Lambda, LangChain, and OpenAI GPT for Computer Vision and Generative AI at Aerolíneas Argentinas. - Built a Post-Call Analytics solution (AWS Transcribe, Comprehend, QuickSight) and deployed Azure Cognitive Search / Form Recognizer for document intelligence at Banco Macro.	
Pi Data Strategy & Consulting , Semi-Senior Data Scientist / Statistician & Economist	Córdoba, Argentina Mar 2021 – Oct 2023
- Built regression, neural network, and multinomial classification models on Azure and Databricks for Anheuser-Busch InBev, Pernod Ricard, and GDM Don Mario. - Applied NLP and Speech Technology (IBM Watson) and sales-forecasting models over georeferenced datasets.	

National Institute of Agriculture Technology (INTA), Lead Statistician / Head of Statistical Research

Córdoba, Argentina
Jan 2012 – Dec 2021

- Led statistical research nationwide: designed experimental designs and mixed/biometric models for ecophysiology, plant breeding, and environmental sustainability programs.
- Applied neural networks, deep learning, and computer vision to UAV and precision-agriculture data for crop health, soil analysis, and pest detection.
- Authored and co-authored 15+ peer-reviewed statistical research publications in applied genomics and agronomy.

Education

Postgraduate Master's National University of Córdoba, Applied Statistics Degree

Córdoba, Argentina
Jan 2006 – Dec 2008

Diploma National University of Córdoba, Data Science, Machine Learning and Applications

Córdoba, Argentina
Jan 2019 – Dec 2020

Bachelor's degree National University of Córdoba, Economics

Córdoba, Argentina
Jan 1993 – Dec 1998

Skills

Agentic & Generative AI: Multi-Agent Systems, LangGraph, LangChain, LlamaIndex, AutoGen, Semantic Kernel, RAG, MCP, Prompt Engineering, LLMs (Gemini, GPT, Claude, Ollama, Hugging Face), AI Governance, Explainable AI (XAI, SHAP).

Machine Learning & Statistics: Deep Learning, Neural Networks, XGBoost, Random Forest, Isolation Forest, Decision Trees, Computer Vision, NLP, Causal Inference, Bayesian Statistics, Econometrics, Anomaly Detection, Time Series Analysis, Experimental Design, Biometrics.

Cloud & Data Engineering: AWS (SageMaker, Bedrock, Lambda, Rekognition, Fraud Detector), Azure AI Services, Databricks, Snowflake, Apache Spark, PySpark, Amazon Athena, Teradata, n8n, Docker, SQL, MLOps.

FDE & Platform Engineering: FastAPI, React, TypeScript, Node.js/Express, Docker, MLflow (PyFunc, Model Serving), Databricks Vector Search & Delta Lake, AWS Lambda, CI/CD, Git, Prometheus, Grafana, OpenTelemetry, Amazon CloudWatch.

Programming & Tools: Python, R, SAS, SQL, Excel, Power BI.

Languages: Spanish (Native), English (B2 - Professional Working Proficiency), Portuguese (Basic).

Certifications

- Databricks Certified Generative AI Engineer Associate — Issued 07/2026, valid through 07/2028 (Credential ID 186901264)
- Databricks Academy — Generative AI Application Deployment & Monitoring, Evaluation & Governance, Building Retrieval/Single-Agent Applications, AI Agent Fundamentals, Unity Catalog Data Governance (2025–2026)
- Generative AI with Large Language Models — DeepLearning.AI & AWS, via Coursera (2023)
- Agentic AI on AWS — BeSA (Become a Solution Architect), Candidate (2026)
- Microsoft Certified: Azure AI Fundamentals
- AWS Academy Graduate — Cloud Foundations (2024) & Machine Learning Foundations (2020)
- MLOps Fundamentals CI/CD/CT pipelines with Azure
- Machine Learning by Professor Andrew Ng, Stanford University / Coursera
- IBM SkillsBuild: Mastering the Art of Prompting — Effective Instructions for AI Language Models